Assignment for Programming for Problem Solving Using C (ESCS291)

Getting Started

1. Write a program to print "Hello, World!".
2. Write a program to add two numbers.
3. Write a program to calculate the area of a circle.
4. Write a program to convert temperature from Fahrenheit to Celsius.
5. Write a program to calculate the perimeter of a rectangle.
6. Write a program to calculate the volume of a cube.
7. Write a program to convert kilometers to meters.
8. Write a program to calculate the area of a triangle using Heron's formula.

C Instructions

9. Write a program to find the sum and average of three numbers.
10. Write a program to swap two numbers using a temporary variable.
11. Write a program to swap two numbers without using a temporary variable.
12. Write a program to calculate simple interest.
13. Write a program to calculate compound interest.
14. Write a program to calculate the area of a rectangle.
15. Write a program to calculate the area of a square.
16. Write a program to calculate the area of a trapezium.
17. Write a program to calculate the area of a rhombus.

Decision Control Instructions

18. Write a program to check if a number is even or odd.
19. Write a program to find the largest of three numbers.
20. Write a program to check if a year is a leap year.
21. Write a program to check if a number is positive, negative, or zero.
22. Write a program to solve a quadratic equation.
23. Write a program to check if a triangle is equilateral, isosceles, or scalene.
24. Write a program to check if a number is divisible by 5 and 11.
25. Write a program to check if a number is a two-digit number.
26. Write a program to check if a number is a three-digit number.

More Complex Decision Making

27. Write a program to check if a number is divisible by 5 and 11.
28. Write a program to check if a triangle is valid (given its sides).
29. Write a program to check if a character is a vowel or consonant.
30. Write a program to calculate the grade of a student based on marks.
31. Write a program to check if a number is a multiple of 3 and 5.
32. Write a program to check if a number is a perfect square.
33. Write a program to check if a number is a perfect cube.
34. Write a program to check if a number is a prime number.
35. Write a program to check if a number is a palindrome.

Loop Control Instructions

36. Write a program to print the first 10 natural numbers.
37. Write a program to print the first 10 natural numbers in reverse order.
38. Write a program to print the multiplication table of a number.
39. Write a program to find the factorial of a number.
40. Write a program to print the Fibonacci series up to a given number.
41. Write a program to print all even numbers between 1 and 100.
42. Write a program to print all odd numbers between 1 and 100.
43. Write a program to print the sum of the first 100 natural numbers.
44. Write a program to print the sum of the first 100 even numbers.
45. Write a program to print the sum of the first 100 odd numbers.

More Complex Repetitions

46. Write a program to check if a number is a palindrome.
47. Write a program to check if a number is an Armstrong number.
48. Write a program to find the sum of digits of a number.
49. Write a program to reverse a number.
50. Write a program to print the first 20 prime numbers.
51. Write a program to print the first 20 Fibonacci numbers.
52. Write a program to print the first 10 Armstrong numbers.
53. Write a program to print the first 10 palindrome numbers.

Case Control Instructions

54. Write a program to implement a simple calculator using switch-case.
55. Write a program to print the day of the week based on a number (1-7).
56. Write a program to check if a character is a vowel using switch-case.
57. Write a program to print the number of days in a month using switch-case.
58. Write a program to print the season based on the month using switch-case.
59. Write a program to print the name of the month based on a number (1-12).
60. Write a Program to Find Grade of a Student Using Switch Case.

Functions

61. Write a program to calculate the factorial of a number using a function.
62. Write a program to check if a number is prime using a function.
63. Write a program to find the GCD of two numbers using a function.
64. Write a program to print the Fibonacci series using recursion.
65. Write a program to calculate the power of a number using a function.
66. Write a program to calculate the sum of digits of a number using a function.
67. Write a program to reverse a number using a function.
68. Write a program to check if a number is a palindrome using a function.
69. Write a program to check if a number is an Armstrong number using a function.

Pointers

70. Write a program to swap two numbers using pointers.
71. Write a program to reverse an array using pointers.

72. Write a program to find the length of a string using pointers.
73. Write a program to sort an array using pointers.
74. Write a program to find the largest element in an array using pointers.
75. Write a program to find the smallest element in an array using pointers.
76. Write a program to find the sum of elements in an array using pointers.
77. Write a program to find the average of elements in an array using pointers.
78. Write a program to find the sum of two matrices using pointers.

Recursion

79. Write a program to solve the Tower of Hanoi problem.
80. Write a program to find the factorial of a number using recursion.
81. Write a program to find the GCD of two numbers using recursion.
82. Write a program to print the Fibonacci series using recursion.
83. Write a program to find the sum of digits of a number using recursion.
84. Write a program to reverse a number using recursion.
85. Write a program to check if a number is a palindrome using recursion.
86. Write a program to check if a number is an Armstrong number using recursion.
87. Write a program to find the power of a number using recursion.

Data Types Revisited

88. Write a program to demonstrate the use of enum.
89. Write a program to demonstrate the use of typedef.
90. Write a program to demonstrate the use of const keyword.
91. Write a program to demonstrate the use of volatile keyword.
92. Write a program to demonstrate the use of static keyword.
93. Write a program to demonstrate the use of extern keyword.
94. Write a program to demonstrate the use of register keyword.
95. Write a program to demonstrate the use of auto keyword.

The C Preprocessor

96. Write a program to demonstrate the use of #define for constants.
97. Write a program to demonstrate the use of #include.
98. Write a program to demonstrate conditional compilation using #ifdef.
99. Write a program to demonstrate the use of #if, #elif, and #else.
100. Write a program to demonstrate the use of #pragma.

Arrays

101. Write a program to find the largest element in an array.
102. Write a program to reverse an array.
103. Write a program to find the sum of elements in an array.
104. Write a program to sort an array using bubble sort.
105. Write a program to search for an element in an array using linear search.
106. Write a program to search for an element in an array using binary search.
107. Write a program to find the second largest element in an array.
108. Write a program to find the second smallest element in an array.

109. Write a program to find the frequency of each element in an array.
110. Write a program to merge two arrays.

Multidimensional Arrays

111. Write a program to add two matrices.
112. Write a program to multiply two matrices.
113. Write a program to find the transpose of a matrix.
114. Write a program to check if a matrix is symmetric.
115. Write a program to find the sum of diagonal elements of a matrix.
116. Write a program to find the product of diagonal elements of a matrix.
117. Write a program to find the sum of each row and column of a matrix.
118. Write a program to find the sum of two matrices.
119. Write a program to find the difference of two matrices.
120. Write a program to find the product of two matrices.

Strings

121. Write a program to find the length of a string.
122. Write a program to concatenate two strings.
123. Write a program to compare two strings.
124. Write a program to reverse a string.
125. Write a program to check if a string is a palindrome.
126. Write a program to count the number of vowels in a string.
127. Write a program to count the number of consonants in a string.
128. Write a program to count the number of words in a string.
129. Write a program to count the number of lines in a string.
130. Write a program to count the number of digits in a string.

Handling Multiple Strings

131. Write a program to sort an array of strings.
132. Write a program to search for a string in an array of strings.
133. Write a program to concatenate multiple strings.
134. Write a program to find the longest string in an array of strings.
135. Write a program to find the shortest string in an array of strings.
136. Write a program to find the frequency of each string in an array of strings.
137. Write a program to find the number of occurrences of a substring in a string.
138. Write a program to replace a substring in a string.
139. Write a program to remove a substring from a string.
140. Write a program to insert a substring into a string.

Structures

141. Write a program to store and display student information using structures.
142. Write a program to store and display employee details using structures.
143. Write a program to demonstrate nested structures -
144. Write a program in C to store and display details of a student where each student has a nested structure for address (street, city, PIN).

145. Write a program to demonstrate the use of arrays of structures –
146. Write a program in C to store details of 5 students (name, roll, marks) using an array of structures and display the highest scorer.
147. Write a program to demonstrate the use of pointers to structures –
148. Write a program in C to access and modify student details using a pointer to structure.
149. Write a program to demonstrate the use of structures with functions –
150. Write a program in C to pass a structure (student details) to a function and display the data using that function.
151. Write a program to demonstrate the use of structures with arrays –
152. Write a program in C to store marks of a student in an array within a structure and calculate total and average marks.
153. Write a program to demonstrate the use of structures with pointers –
154. Write a program in C to dynamically allocate memory for a structure using pointers and store employee details.
155. Write a program to demonstrate the use of structures with nested structures
156. Write a program in C to store employee details where each employee has a nested structure for date of joining (day, month, year).
157. Write a program to demonstrate the use of structures with unions –
158. Write a program in C to store student information using a structure that contains a union for either marks or grade, depending on user choice.

Console Input/Output

159. Write a program in C to read characters one by one using getchar() until newline is encountered and display them in uppercase using putchar().
160. Write a program in C to read a string using gets() and display the string in reverse order using puts().
161. Write a program in C to read student details (name, roll, marks) using scanf() and display them in a formatted table using printf().
162. Write a program in C to read a line of text using fgets() and write it to a file using fputs().
163. Write a program in C to copy contents of one file to another character by character using fgetc() and fputc().
164. Write a program in C to read structured data (name and marks) from a file using fscanf() and write the formatted result to another file using fprintf().
165. Write a program in C to store formatted data into a string using sprintf() and then extract values from the string using sscanf().
166. Write a program in C to write an array of structures (student records) to a binary file using fwrite() and read them back using fread().
167. Write a program in C to determine the size of a file and reposition the file pointer using fseek() and ftell().
168. Write a program in C to read a file, display its contents, then use rewind() to reset the file pointer and display the contents again.

File Input/Output

169. Write a program to create a file and write data to it.
170. Write a program to read data from a file and display it.
171. Write a program to copy the contents of one file to another.
172. Write a program to count the number of lines, words, and characters in a file.
173. Write a program to search for a specific word in a file.
174. Write a program to replace a specific word in a file.
175. Write a program to append data to a file.
176. Write a program to delete a specific line from a file.
177. Write a program to insert a specific line into a file.
178. Write a program to sort the contents of a file.

More Issues in Input/Output

179. Write a program in C to determine the size of a file and reposition the file pointer using fseek() and ftell().
180. Write a program in C to write and read an array of structures (student records) to/from a binary file using fwrite() and fread().
181. Write a program in C to store the current file position using fgetpos() and later restore it using fsetpos() while reading a file.
182. Write a program in C to read a file and demonstrate end-of-file detection using feof() and error handling using ferror().
183. Write a program in C to demonstrate clearing file error and EOF indicators using clearerr() after an error occurs during file reading.
184. Write a program in C to open a file and display appropriate error messages using perror() if the file cannot be accessed.
185. Write a program in C to create a temporary file using tmpfile(), write data to it, and read the data back.
186. Write a program in C to generate a temporary filename using tmpnam(), create a file with that name, and write data into it.
187. Write a program in C to delete a file using remove() after confirming its existence.
188. Write a program in C to rename an existing file using rename() and verify the change.

Operations on Bits

189. Write a program to demonstrate bitwise operators.
190. Write a program to check if a number is even or odd using bitwise operators.
191. Write a program to swap two numbers using bitwise operators.
192. Write a program to find the number of set bits in a number.
193. Write a program to find the number of unset bits in a number.
194. Write a program to find the number of trailing zeros in a number.
195. Write a program to find the number of leading zeros in a number.
196. Write a program to find the number of trailing ones in a number.
197. Write a program to find the number of leading ones in a number.

198. Write a program to find the number of set bits in a number using Brian Kernighan's algorithm.

Miscellaneous Features

199. Write a program in C to accept and display command-line arguments and count the total number of arguments passed.
200. Write a program in C to dynamically allocate memory for an array of integers using malloc(), accept values from the user, and display them.
201. Write a program in C to dynamically allocate memory for an array using calloc(), initialize elements, and display the values.
202. Write a program in C to dynamically resize an array using realloc() and demonstrate addition of new elements.
203. Write a program in C to allocate memory dynamically, use it, and properly release it using free().
204. Write a program in C to register multiple functions using atexit() and demonstrate their execution upon normal program termination.
205. Write a program in C to demonstrate non-local jumps using setjmp() and longjmp() for error handling.
206. Write a program in C to handle interrupt signals (e.g., SIGINT) using signal() and display a custom message when triggered.
207. Write a program in C to raise a signal programmatically using raise() and handle it using a signal handler.
208. Write a program in C to demonstrate abnormal program termination using abort() and observe its effect.